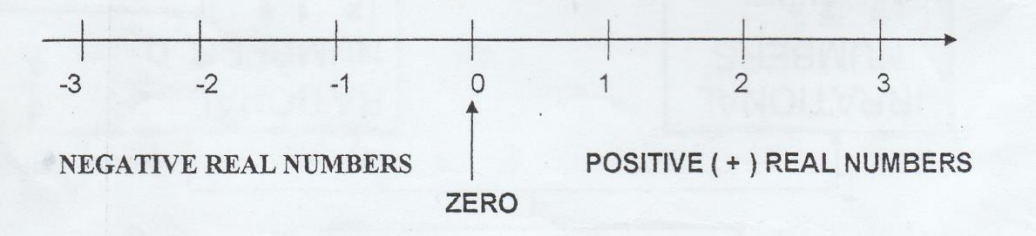
Chapter 1

1.1 Inequalities

Real number line

The real numbers can be represented by points on a line. It consists of 3 classes of real numbers, namely, negative real numbers, zero and positive real numbers as shown below.



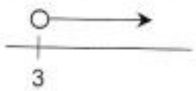
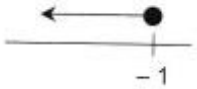
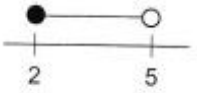
Definition of Inequalities

An inequality is a mathematical sentence formed by relating two expressions to each other. The symbols of inequalities and their respective meanings are as follows.

Symbols	Interpretations
>	Greater than
≥	Greater than or equal to
<	Less than
≤	Less than or equal to

Intervals

There are 3 ways to represent intervals, namely Inequality Form, Interval Notation and Real Number Line (RNL).

Inequality Form	Interval Form	Real Number line
$x > 3$	$(3, \infty)$	
$x \leq -1$	$(-\infty, -1]$	
$2 \leq x < 5$	$[2, 5)$	

Try out!

Find the interval notation and RNL of the following inequalities.

a) $x < 3$
b) $x \geq -2$
c) $2 < x \leq 5$

Solving Inequalities

The problem of inequalities can be categorized into 3 types. That is Linear Inequalities, Quadratic Inequalities and Rational Inequalities.

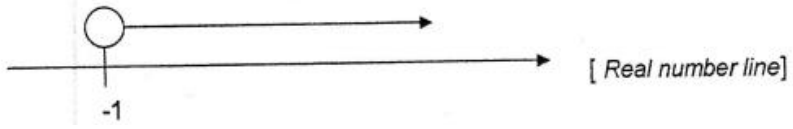
- **Linear Inequalities**

Example 1
Solve the following inequality and express your answer in interval notation and RNL.

$$3 - 2x < 5$$

Solution:
$3 - 2x - 3 < 5 - 3$
$-2x < 2$
$\frac{-2x}{-2} > \frac{2}{-2}$
$x > -1$

The answer is $x = (-1, \infty)$ *[interval notation]* or $\{x \in R, x > -1\}$ *[set notation]*



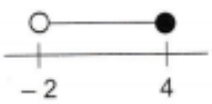
Example 2
Solve the following inequality and express your answer in interval notation and RNL.

$$-3 < \frac{4 + 2x}{2} - 3 \leq 3$$

Solution

$(+3)$: $0 < \frac{4 + 2x}{2} \leq 6$
$(\times 2)$: $0 < 4 + 2x \leq 12$
(-4): $-4 < 2x \leq 8$
$(\div 2)$: $-2 < x \leq 4$

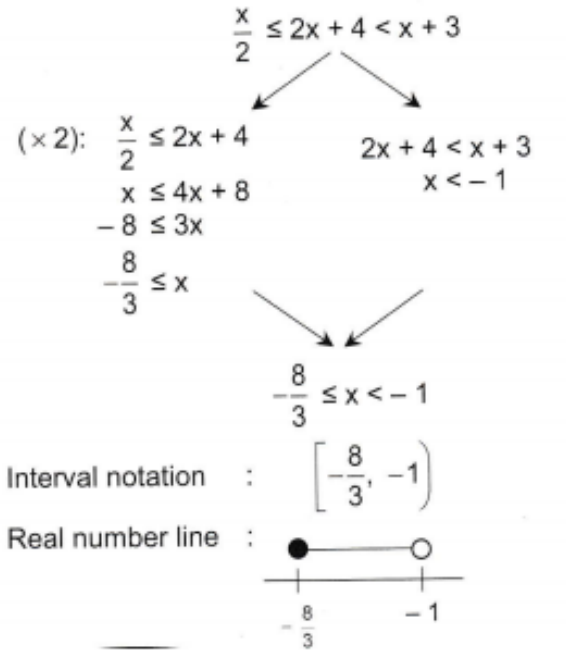
Interval notation : $(-2, 4]$
Real number line :



Example 3
Solve the following inequality and express your answer in interval notation and RNL.

$$\frac{x}{2} \leq 2x + 4 < x + 3$$

Solution



Reminder

Multiplication or **Division** of a negative value will change the inequality sign.

- **Quadratic Inequalities**

They must be solved by using **Interval Table**.

Example 4
Solve the following inequality and express your answer in interval notation and RNL.

$$(3x+2)^2 < 144x^2 - 288x + 144$$

Solution

$9x^2 + 12x + 4 < 144x^2 - 288x + 144$

$0 < 135x^2 - 300x + 140$

$(\div 5) \quad 27x^2 - 60x + 28 > 0$

$(9x - 14)(3x - 2) > 0$

Critical values: $9x - 14 = 0$; $3x - 2 = 0$

$x = \frac{14}{9}$; $x = \frac{2}{3}$

Critical line:

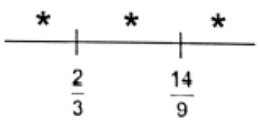
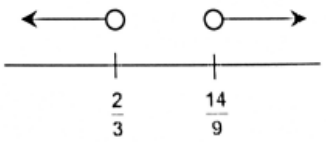


Table:

Interval	Test Value	Sign of $(9x - 14)(3x - 2) > 0$
$\left(-\infty, \frac{2}{3}\right)$	0	$(-)(-) = +$ √
$\left(\frac{2}{3}, \frac{14}{9}\right)$	1	$(-)(+) = -$
$\left(\frac{14}{9}, \infty\right)$	3	$(+)(+) = +$ √

Interval form : $\left(-\infty, \frac{2}{3}\right) \cup \left(\frac{14}{9}, \infty\right)$

Real number line :



- **Rational Inequalities**

Like quadratic inequalities, they must be solved by using **Interval Table**.

Example 5
Solve the inequality below. Illustrate the solution on a real number line.

$$\frac{x+4}{x-2} \geq 2$$

Solution

$$\frac{x+4}{x-2} - 2 \geq 0$$

$$\frac{x+4-2(x-2)}{x-2} \geq 0$$

$$\frac{x+4-2x+4}{x-2} \geq 0$$

$$\frac{8-x}{x-2} \geq 0$$

Critical values: $8 - x = 0$; $x - 2 = 0$

$8 = x$; $x = 2$

Critical line:

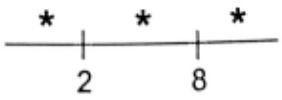


Table:

Interval	Test Value	Sign of $\frac{8-x}{x-2} \geq 0$
$(-\infty, 2)$	0	$\frac{(+)}{(-)} = -$
$(2, 8]$	5	$\frac{(+)}{(+)} = +$ √
$[8, \infty)$	10	$\frac{(-)}{(+)} = -$

Real number line :



Important note
Why $x = 2$ is not included (open bracket)?

Remember the concept of fraction $\frac{a}{b}$ where $b \neq 0$
So, $x - 2 \neq 0$
$x \neq 2$
∴ x is not equal to 2 (2 is not included